

Basem Mohamed

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EDUCATION

University of Houston
Bachelor of Science, Computer Science

Aug 2023 – May 2027
Houston, TX

PROJECTS

AI Room Remodeler (Hackathon Winner) | *Express.js, TailwindCSS, SerpAPI, Gemini* **Oct 2025**

- Earned 2nd Place by CodeRED and Best Use of Gemini API by Major League Hacking.
- Built a full-stack prototype that turns a room photo into redesign options plus a shoppable furniture list with live product sourcing.
- Integrated Google Gemini 2.5 Flash-Image & Pro models for image-to-image generation and JSON-based furniture metadata extraction.
- Implemented SerpAPI's Google Shopping Scraper to turn Gemini outputs into product listing links.

Dream Maker (Image Generator with Daily Quotas) | *React, Node.js, SQLite*, **May – Sept 2025**

- Built a React web app that turns user provided descriptions of dreams into images using a React/Vite front end and an Express API.
- Added cost/abuse controls by issuing anonymous user IDs (cookie-based) and enforcing 20 generations per 24 hours + UI countdown showing remaining usage.
- Improved reliability and API hygiene with server-side validation, clear HTTP errors (400/403/500), and atomic quota updates in SQLite to prevent counter desync.
- Deployed the service (front end + API) and configured persistent local storage for usage tracking.

The Scholar's Companion | *Flask, Google Search API, Bootstrap, C++* **May 2023 – June 2024**

- Developed a tool that ranks Google results by reading difficulty for non-native English speakers.
- Designed a readability-scoring algorithm grounded in established linguistic research to classify educational text from grade-school through college level.
- Collaborated in a two-person team to deploy and maintain the application for over 12 months, supporting ~50 monthly active users and resolving user-reported issues.

WORK EXPERIENCE

Freelance Software Developer **Feb 2026 – Present**
E-commerce Freight & Shipping System / WooCommerce, FedEx API *Houston, TX*

- Designed a hybrid shipping system routing orders between live FedEx rates and a custom freight workflow, ensuring accurate pricing for all products
- Eliminated manual shipping operations by automating rate calculation based on customer address and order details, and replacing freight handling with a scalable, on-demand quote system

Multi-Level Math Instructor **Oct. 2023 – Sept. 2024**
Mathnasium *Houston, TX*

- Taught and coached 40+ students across multiple grade levels by translating complex ideas into simple steps and adjusting explanations in real time.
- Diagnosed learning gaps through short assessments and observation and created targeted practice plans that improved student confidence and consistency.
- Communicated progress and concerns clearly with parents and staff and balanced fast-paced peak hours with accuracy and professionalism

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, HTML, CSS

Frameworks: React, Node.js, Express, Flask, Vite

Databases: SQLite, PostgreSQL

Developer Tools: Git, VS Code, ESLint

Spoken Languages: English (Fluent), Arabic (Fluent)